

HU Collection

TMUA Mock Set G Paper 2

Worked Solutions

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Answer key

Q	1	2	3	4	5	6	7	8	9	10
Ans	D	B	C	A	E	B	F	D	F	C
Q	11	12	13	14	15	16	17	18	19	20
Ans	A	E	D	B	E	C	A	E	D	B

Each solution gives the intended route and names the trap the question was built around. Questions are ordered by increasing difficulty.

Question 1. Answer D.

Check both values directly. $x = 3$: $\sqrt{9} - \sqrt{1} = 3 - 1 = 2$. $x = 11$: $\sqrt{25} - \sqrt{9} = 5 - 3 = 2$. Both work. Squaring can introduce false solutions, but that is a reason to *check* candidates, not to reject one by size. Here both sides were non-negative at each squaring for both values, so nothing extraneous appeared.

Question 2. Answer B.

With C false, A, B and D are all true. B makes $n + 1$ a square: $n \in \{3, 8, 15, 24, 35, 48, 63, 80, 99\}$. D keeps only those with digit sum 8: 8, 35 and 80. A: 35 is divisible by neither 2 nor 3, so it fails. 8 satisfies A, but it has exactly four divisors (1, 2, 4, 8), so C would be true. 80 satisfies A, and it has ten divisors, so C is false as required. So $n = 80$.

Question 3. Answer C.

I: a^2 divides $b^2c^2 = (bc)^2$, and comparing the power of each prime shows $a \mid bc$. II fails: $6 \mid 2 \cdot 3$ but $6 \nmid 2$ and $6 \nmid 3$. III: $b = a\beta$ and $c = a\gamma$ give $bc = a^2\beta\gamma$. IV: each prime occurs in a^2 to twice the power it has in a , so $a^2 \mid b^2$ forces the powers in a to be at most those in b . So I, III and IV. Statement I is the one that looks false and is not.

Question 4. Answer A.

I: $(a + b)^2 = 1 + 2ab \leq 1 + (a^2 + b^2) = 2$. III: $(a + b)^2 \geq 0$ gives $2ab \geq -1$. II: $|a|, |b| \leq 1$, so $a^3 \leq a^2$ and $b^3 \leq b^2$, giving $a^3 + b^3 \leq a^2 + b^2 = 1$. IV fails at $a = b = -\frac{1}{\sqrt{2}}$, where $a + b = -\sqrt{2}$ but $ab = \frac{1}{2}$. So I, II and III.

Question 5. Answer E.

I and II follow from the triangle inequality: the distances from x to 1 and to -1 sum to at least 2 and differ by at most 2. III: if $|x| < 1$ the left side is non-negative and the right side negative; if $|x| \geq 1$ it reads $x^2 \geq |x|$, which holds. IV: a sum of distances to 2, 5, 9 is least at the median 5, where it is $3 + 0 + 4 = 7$. V fails whenever $|x| < 3$, since the left side is positive and the right side negative. So I to IV.

Question 6. Answer B.

I is the standard argument and is valid. II is less familiar but also complete: the case p even forces a common factor 2, and the case p odd is ruled out by squares modulo 8. III is invalid: $\frac{1}{3} = 0.333\dots$ has a non-terminating decimal expansion and is rational. What characterises rational numbers is a decimal that eventually repeats, and the argument does not show that this one never does.

Question 7. Answer F.

IV is the definition in prime-factor form. I: divisors pair up as d and $\frac{n}{d}$, and a divisor is unpaired exactly when $d = \sqrt{n}$. II is necessary but not sufficient: 5 leaves remainder 1 and is not a square. III fails both ways: 2 has divisor sum 3, which is odd, although 2 is not a square. (The divisor sum is odd exactly for squares and twice squares.) So I and IV.

Question 8. Answer D.

I: positive for all $n > 100$. II: for odd $n \geq 3$ the terms are $-1 + \frac{2}{n} < 0$, and odd n occur beyond every N , so no. III: $\sin n \geq -1$, so every term is at least $\frac{1}{2}$: yes, with $N = 0$. IV: positive for $n > 1000$. V: when $n \equiv 2 \pmod{4}$, $\cos \frac{n\pi}{2} = -1$ and the term is 0, which is not positive, and such n occur beyond every N , so no. Three are eventually positive. Note that V is never negative; “eventually positive” needs strictly positive terms.

Question 9. Answer F.

By Vieta, $-a$ is the sum of the roots, b the sum of products in pairs, $-c$ the sum of products in threes and d the product. With all roots positive, I and II follow. III: $a^2 - 2b$ is the sum of the squares of the roots, which is positive. IV can fail when the roots are close together: 1, 1.01, 1.02, 1.03 give $a^2 \approx 16.48$ and $3b \approx 18.54$. (The sharp inequality is $3a^2 \geq 8b$.) So I, II and III.

Question 10. Answer C.

Line IV, $3k^3 \geq (k+1)^3$, is equivalent to $\sqrt[3]{3}k \geq k+1$, which holds only for $k \geq \frac{1}{\sqrt[3]{3}-1} \approx 2.26$, that is for $k \geq 3$. It fails at $k = 1$ ($3 < 8$) and $k = 2$ ($24 < 27$), so the chain from $n = 1$ breaks immediately. Yet the claim is true: $n = 1, 2, 3$ can be checked directly ($3 \geq 1$, $9 \geq 8$, $27 \geq 27$), and the step works from $k = 3$ onwards. A correct proof needs those three starting cases. Option F would still leave the step unjustified at $k = 2$.

Question 11. Answer A.

I: f falls continuously from 2 to -2 on $[1, 3]$. IV: f is increasing on $(-\infty, 1]$, so $f(0) < f(1) = 2$. II and III depend on behaviour at the ends that is not given. f might increase towards a limit such as 1 as $x \rightarrow -\infty$, and never be 0 there; and for $x > 3$ it might increase towards 0 or towards 1, never reaching 2. So there could be as few as one solution of $f(x) = 0$ and as few as one of $f(x) = 2$. Only I and IV are forced.

Question 12. Answer E.

I: the regions under f and under f^{-1} (reflected) fill the unit square, so the two integrals sum to 1. II: if $f(x) \geq x$ throughout, the integral would be at least $\frac{1}{2}$. III and IV can fail. The piecewise-linear function through $(0, 0)$, $(\frac{2}{5}, \frac{1}{50})$, $(\frac{1}{2}, \frac{13}{25})$, $(\frac{9}{10}, \frac{89}{150})$ and $(1, 1)$ is increasing, and adding up the trapezia gives an integral of exactly $\frac{1}{3}$, yet $f(\frac{1}{2}) = \frac{13}{25} > \frac{1}{2}$. Keeping f tiny on most of $[0, \frac{1}{2}]$ pays for the jump. So I and II.

Question 13. Answer D.

I makes f positive everywhere. IV: $c = f(0) > 0$ and $a + b + c = f(1) > 0$. If $a > 0$ the vertex $-\frac{b}{2a} \leq 0$, so f is increasing on $[0, 1]$ and its minimum there is $f(0) > 0$. If $a \leq 0$ the graph is a downward parabola or a line, whose minimum on an interval is at an endpoint. Either way $f > 0$ on $[0, 1]$. II and III fail: $f(x) = 100\left(x - \frac{1}{4}\right)^2 - 1$ has $f(0) = f\left(\frac{1}{2}\right) = 5.25$ and $f(1) = 55.25$, but $f\left(\frac{1}{4}\right) = -1$. Sampling points can never rule out a narrow dip.

Question 14. Answer B.

Let $g(x) = x^3 + px$. For $p < 0$ it has stationary values $\pm 2\left(-\frac{p}{3}\right)^{3/2}$, and the equation $g(x) = -q$ has three distinct roots exactly when $|q| < 2\left(-\frac{p}{3}\right)^{3/2}$. II: $p < -3$ makes that bound exceed 2, so $|q| < 2$ suffices. III: the roots are 0 and $\pm\sqrt{-p}$. I and IV fail for $p = -3$, $q = 5$: the bound is 2, and $x^3 - 3x + 5$ has only one real root.

Question 15. Answer E.

The AM-GM step is correct, but its right-hand side is not a constant, and $\frac{2}{\sqrt{x}} \geq 2$ holds only for $x \leq 1$. The claim itself is false. The minimum of $x + \frac{1}{x^2}$ is at $1 - \frac{2}{x^3} = 0$, that is $x = \sqrt[3]{2}$, where it equals $\sqrt[3]{2} + \frac{1}{\sqrt[3]{4}} = \frac{3}{\sqrt[3]{4}} \approx 1.89 < 2$. For instance $x = 1.2$ gives about 1.894. AM-GM gives a useful bound only when the product of the terms is constant: here, split x as $\frac{x}{2} + \frac{x}{2}$.

Question 16. Answer C.

I: a function increasing throughout $(0, 1)$ cannot return to its starting value, so no. II: a function decreasing throughout $(0, 1)$ cannot go from 0 up to 1, so no. IV: a slope never exceeding 1 allows a rise of at most 1 over an interval of length 1, not 2, so no. III: yes. For example $f(x) = 9x - 8x^2$ has $f(0) = 0$ and $f(1) = 1$, and its maximum, at $x = \frac{9}{16}$, is $\frac{81}{32} > 2$, so by continuity it takes the value 2. Only III.

Question 17. Answer A.

List the representable numbers: 5, 7, 10, 12, 14, 15, 17, 19, 20, 21, 22, 24, 25, 26, 27, 28, ... From 24 onwards, 24, 25, 26, 27, 28 are five consecutive representable numbers, so adding 5s covers everything beyond. So I holds. The non-representable numbers are 1, 2, 3, 4, 6, 8, 9, 11, 13, 16, 18, 23: twelve, so II holds. III fails: 9 and 18 are multiples of 3 greater than 6 that are not representable.

Question 18. Answer E.

I: if $f(a) = f(b)$ then $a = f(f(a)) = f(f(b)) = b$. II fails for $f(x) = -x$. III: if $f(x) > x$ for some x , applying the increasing f gives $x = f(f(x)) > f(x)$, a contradiction, and $f(x) < x$ fails in the same way. IV fails for a discontinuous f : let $f(x) = x + 1$ when the integer part of x is even and $f(x) = x - 1$ when it is odd. This swaps $[0, 1)$ with $[1, 2)$, $[2, 3)$ with $[3, 4)$, and so on, so $f(f(x)) = x$ but $f(x) \neq x$ everywhere. For a continuous f , IV would hold.

Question 19. Answer D.

I: $f(x) = e^x$. II: $f(x) = e^{-x}$. III is impossible: $f'' < 0$ makes f' decreasing, so for $x > 0$, $f'(x) < f'(0) < 0$ and $f(x) < f(0) + f'(0)x$, which becomes negative for large x . IV is impossible for the mirror-image reason: f' decreasing means $f'(x) > f'(0) > 0$ for $x < 0$, so $f(x) < f(0) + f'(0)x$,

which becomes negative for large negative x . A graph that bends downwards must eventually fall below any line it is heading along, in one direction or the other.

Question 20. Answer B.

II is immediate. If $|x| > 2$ then $|f(x)| = x^2 - 2 > |x|$, so the iterates grow in size and never return: every solution of any $f^{(n)}(x) = x$ lies in $[-2, 2]$, and IV fails. On $[-2, 2]$, f runs down from 2 to -2 and back up to 2, covering $[-2, 2]$ twice. So $f^{(n)}$ covers $[-2, 2]$ exactly 2^n times in 2^n monotonic sweeps between -2 and 2 , and the line $y = x$, which runs from corner to corner of the square, crosses each sweep exactly once. That gives 4 solutions for $f^{(2)}$ and 8 for $f^{(3)}$, and every solution is real. I and III hold.